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Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
B.Tech Degree S7 (R, S) Examination November 2024 (2019 Scheme)

Course Code: EET413
Course Name: ELECTRIC DRIVES

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

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| 1 | Classify the power modulators used in an electric drive. Give a brief description of each type. | (3) |
| 2 | Comment on the nature of load torque in i) Paper mill drives ii) Fan, compressors, etc. Also, give the classification of load torque. | (3) |
| 3 | Calculate the voltage supplied by a single-phase full converter operated with a firing angle of 60 degrees, when it is supplied by a 230 V, 50 Hz AC source. | (3) |
| 4 | Draw the circuit diagram of a 3-phase semi-converter fed separately excited DC motor drive. | (3) |
| 5 | Describe the Regenerative braking control of a separately excited DC motor using a single quadrant chopper. | (3) |
| 6 | Explain how motoring and regenerative braking are achieved in two quadrant chopper control of a DC series motor. | (3) |
| 7 | With a neat diagram, explain the stator voltage control method for speed control of induction motors. What is the limitation? | (3) |
| 8 | Why V/f control method is considered superior to the frequency control? Explain the method with a block diagram. | (3) |
| 9 | What is the necessity of coordinate transforms? Describe Park's transformation. | (3) |
| 10 | Explain Clark's transformation with necessary equations. | (3) |

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

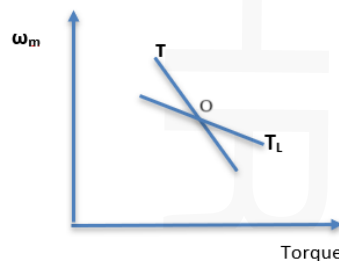
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| 11 | a) Explain the dynamics of a motor-load system. Elaborate on the components of load torque. | (7) |
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- b) A rotational load is connected to the motor through a gear in addition to a rotational load directly coupled to the shaft. Show the effect of gearing on the moment of inertia and load torque. (7)

OR

- 12 a) What is an Electric drive? Explain the advantages and disadvantages of an electric drive. (7)
- b) Derive the condition for steady-state stability. Analyze the steady-state stability of the operating point O shown in Figure. (7)



Module II

- 13 a) Explain the control of a DC separately excited DC motor operated from a 1-phase fully controlled converter. Give necessary circuits and waveforms. (7)
- b) A 220V, 960 rpm, 12.8 A separately excited dc motor has armature circuit resistance and inductance of 2 ohm and 150 mH respectively. It is fed from a single-phase half-controlled rectifier with an ac source voltage of 230V, 50 Hz. Calculate motor torque for $\alpha = 60^\circ$ and speed 600 rpm. Assume continuous conduction. (7)

OR

- 14 a) Why a single-phase semi converter can deliver only single-quadrant operation? Explain the operation with a circuit diagram and necessary waveforms. (7)
- b) A 200 V, 875 rpm, 150 A separately excited DC motor has an armature resistance of 0.06Ω . It is fed from a 1-phase fully controlled rectifier with an AC source voltage of 220 V, 50 Hz. Assuming continuous conduction, Calculate, (7)
- 1) Firing angle for rated motor torque & 750 rpm
 - 2) Firing angle for rated motor torque & -500 rpm

Module III

- 15 a) Explain with a block diagram, the closed-loop control scheme for separately excited DC motor. (7)

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- b) Which are the different chopper circuits used for the control of a DC series motor? (7)
Explain any two circuits in detail.

OR

- 16 With a neat circuit diagram explain the four-quadrant chopper drive for a DC motor. (14)

Module IV

- 17 a) Explain VSI control of an Induction motor using sine triangle PWM. (7)
b) With a block diagram, describe the closed-loop speed control of the induction motor using static rotor resistance control. (7)

OR

- 18 a) A 3-phase 400V, 6-pole 50Hz Δ connected wound rotor induction motor has rotor resistance of 0.2Ω and leakage reactance of $1\Omega/\text{phase}$ referred to stator. When driving a fan load, it runs at full load at 4% slip. What resistance must be inserted in the rotor circuit to obtain a speed of 850rpm? Neglect stator impedance and magnetizing branch. Stator to rotor turns ratio is 2.2. (7)
b) Explain a slip power recovery scheme where speed control only below synchronous speed is achieved. (7)

Module V

- 19 Explain the different modes of variable frequency control of a synchronous motor drive. Compare and contrast the true synchronous mode and self-controlled mode of operation of synchronous motors. (14)

OR

- 20 a) List the advantages of field-oriented control. (4)
b) Explain the direct vector control of the induction motor with suitable diagrams. (10)
